

课程详述

COURSE SPECIFICATION

以下课程信息可能根据实际授课需要或在课程检讨之后产生变动。如对课程有任何疑问，请联系授课教师。

The course information as follows may be subject to change, either during the session because of unforeseen circumstances, or following review of the course at the end of the session. Queries about the course should be directed to the course instructor.

1.	课程名称 Course Title	概率论与数理统计 Probability and Statistics
2.	授课院系 Originating Department	数学系 Department of Mathematics
3.	课程编号 Course Code	MA212
4.	课程学分 Credit Value	3
5.	课程类别 Course Type	专业基础课 Major Foundational Courses
6.	授课学期 Semester	春季 Spring / 秋季 Fall
7.	授课语言 Teaching Language	英文 English / 中英双语 English & Chinese
8.	授课教师、所属学系、联系方式 (如属团队授课, 请列明其他授课教师) Instructor(s), Affiliation & Contact (For team teaching, please list all instructors)	温家强, 数学系 理学院 M607 wenjq@sustech.edu.cn 15169096627 Jiaqiang WEN, Department of Mathematics Rm. 607, College of Science wenjq@sustech.edu.cn 15169096627
9.	实验员/助教、所属学系、联系方式 Tutor/TA(s), Contact	王甫东(WANG Fudong) < wangfd3@sustech.edu.cn > 田锋(TIAN Feng) < 12331104@mail.sustech.edu.cn > 杨懿(YANG Yi) < 12432033@mail.sustech.edu.cn >
10.	选课人数限额(可不填) Maximum Enrolment	100

(Optional)

11. 授课方式

Delivery Method

学时数

Credit Hours

讲授 Lectures	习题/辅导/讨论 Tutorials	实验/实习 Lab/Practical	其它(请具体注明) Other (Please specify)	总学时 Total
48	16			

12. 先修课程、其它学习要求
Pre-requisites or Other
Academic Requirements

MA127 高等数学 (下) 或者 MA102a 数学分析 II
Calculus II or Mathematical Analysis II

13. 后续课程、其它学习规划
Courses for which this course
is a pre-requisite

14. 其它要求修读本课程的学系
Cross-listing Dept.

教学大纲及教学日历 SYLLABUS

15. 教学目标 Course Objectives

介绍概率论和数理统计的基本理论, 基本方法和简单应用, 为更高层次的概率理论及统计理论打下基础。同时着重介绍如何正确地应用统计方法以及对统计意义的正确阐释。本课程的重点是概率和统计方法的应用。

To introduce the basic concepts in probability and statistics which form the basis for all applications of statistics, and for further probability and statistical theory. Also to introduce statistical methods with a strong emphasis on applying standard statistical techniques appropriately and with clear interpretation. The emphasis is on applications.

16. 预达学习成果 Learning Outcomes

通过本课程的学习, 学生将掌握一些基本的概率方法和技术。熟悉能够解决实际生活中带有概率性质的问题的基本方法和技术, 对一系列常用统计方法获得理论联系实践的理解。学生还应对一些统计软件建立初步的了解。

After completing this course, students should understand basic probability methods and techniques. They should be familiar with a range of techniques for solving real life problems of the probabilistic nature and should have a conceptual and practical understanding of a range of commonly applied statistical procedures. They should have also developed some familiarity with the statistical package.

17. 课程内容及教学日历 (如授课语言以英文为主, 则课程内容介绍可以用英文; 如团队教学或模块教学, 教学日历须注明主讲人)

Course Contents (in Parts/Chapters/Sections/Weeks. Please notify name of instructor for course section(s), if this is a team teaching or module course.)

第 1 章 概率的基本概念 (7 学时)

样本空间、随机事件、概率测度、计数方法、条件概率、独立事件、贝叶斯公式

Chapter 1 Basic Concepts of Probability (7 hours)

sample spaces, random events, probability measures, probability calculation, conditional probability, independent event, Bayesian formula

第 2 章 随机变量 (7 学时)

离散型随机变量: 概率分布函数, 常见的离散型随机变量, 如伯努利随机变量、二项分布、几何分布、泊松分布、负二项分布、超几何分布等

连续型随机变量: 概率密度函数、常见的连续型随机变量, 如均匀分布、指数分布、正态分布等

Chapter 2 Random variables (7 hours)

discrete random variables: probability distribution function, Bernoulli random variables, Binomial distribution, geometric and negative binomial distribution, hypergeometric distribution, Poisson distribution

Continuous random variables: probability density function, uniform distribution, exponential distribution, normal distribution

第 3 章 联合分布 (8 学时)

随机变量的联合累积分布函数、边缘累积分布函数。

离散型随机变量的联合频率函数、边缘频率函数。

连续型随机变量的联合密度函数, 边缘密度函数。

独立随机变量、条件分布、联合分布的随机变量函数。

Chapter 3 Joint Distributions (8 hours)

Random variable, discrete random variable, continuous random variable, independent random variable, joint distribution

——前三章为期中考试内容——

第 4 章 随机变量的数字特征 Digital characteristics of r.v. (6 学时)

随机变量的期望、方差和标准差、协方差和相关系数、条件期望。

Chapter 4 Expectation (6 hours)

Expected value, variance, standard deviation, correlation, correction coefficient, conditional expected value

第 5 章 极限定理 (2 学时)

大数定律和中心极限定理

Chapter 5 Limit theorems (2 hours)

The law of large numbers, central limit theorem

第 6 章 数理统计的基本概念及抽样分布 (4 学时)

数理统计的基本概念: 总体和样本、统计推断等。

样本分布, 常用统计量。

t 分布, 卡方分布、F 分布。

Chapter 6 Sampling distributions of estimates (4 hours)

Basic concepts of statistics: sample set and sample, statistic assumption

Sample distribution, commonly used statistics, t distribution, chi-square distribution, F distribution

第 7 章 参数估计 (6 学时)

点估计, 区间估计

Chapter 7 Parametric estimation problems (6 hours)

Point estimation and interval estimation

第 8 章 假设检验 (6 学时)

单样本正态总体均值和方差的检验、两独立样本比较、配对样本比较。

Chapter 8 Testing hypothesis (6 hours)

Basic principles, specification of the significance level and the concept of NULL Hypothesis and alternative hypothesis

18. 教材及其它参考资料 Textbook and Supplementary Readings

教材: 数理统计与数据分析, Mathematical Statistics and Data Analysis, John A. Rice, 3rd Ed.

推荐参考书 Reference books:

1. Probability and Statistics, 5th edition, Jay L. Devore

2. 概率论与数理统计, 浙大版

课程评估 ASSESSMENT

19. 评估形式 Type of Assessment	评估时间 Time	占考试总成绩百分比 % of final score	违纪处罚 Penalty	备注 Notes
出勤 Attendance		10		
课堂表现 Class Performance				
小测验 Quiz				
课程项目 Projects				
平时作业 Assignments		10	晚交 1 天按 50%算, 晚交 2 天按 0 分处理	
期中考试 Mid-Term Test		30		期中考试时间为 4 月 19 日(第九周周六)19:00—21:00 无法补考
期末考试 Final Exam		50		
其它 (可根据需要 改写以上评估方式) Others (The above may be modified as necessary)				

20. 记分方式 GRADING SYSTEM

- ☒ A. 十三级等级制 Letter Grading
☐ B. 二级记分制 (通过/不通过) Pass/Fail Grading

课程审批 REVIEW AND APPROVAL

21. 本课程设置已经过以下责任人/委员会审议通过
This Course has been approved by the following person or committee of authority

教学负责人签字:
日期: